Remark 15.7. Injectivity is required so that distinct formulas receive distinct names; without it the reflection sequence of Definition 15.11 could collapse at some finite stage for purely notational reasons, and the hierarchy below would be an artifact of the naming rather than of the machine.

Discussion. Definition 15.5 is the exact analogue for formulas of what Definition 13.3 does for machines. One may speak of the sentence "Picard is human" without first assigning that sentence an integer; one gives it a name and quotes the name.

15.3 The Internal Provability Predicate

Definition 15.8 (Internal provability predicate)**.** An *internal provability predicate* for a tagged ATP M is a binary predicate symbol Pr of L, with intended reading

$$\mathsf{Pr}(\langle M\rangle, \langle\varphi\rangle) \text{ is true iff the machine named } \langle M\rangle \text{ proves the formula named } \langle\varphi\rangle.$$

Remark 15.9 (What Pr is not). Pr is a primitive predicate symbol of the object language, interpreted by the declared naming conventions. It is not an arithmetized provability predicate constructed from a proof relation inside a theory of arithmetic. In particular, the Hilbert–Bernays–Löb derivability conditions are not assumed to hold of it, and nothing in this section presupposes them. Remark 15.31 records what happens if one does strengthen Pr far enough for those conditions to apply.

Remark 15.10 (Soundness is a separate question). Definition 15.8 gives the intended reading of Pr; it does not assert that M proves only true instances of Pr, nor that M proves all true instances. Those are the two independent hypotheses isolated in Definitions 15.21 and 15.26, and the interest of the hierarchy lies precisely in the fact that neither is automatic.

15.4 The Reflection Sequence

Definition 15.11 (Reflection sequence)**.** Let M be a tagged ATP whose language has a formula-naming operator and an internal provability predicate. The *reflection sequence* of M is the sequence of formulas $(\theta_k)_{k\geq 0}$ defined by

$$\theta_0 := \mathsf{ATP}(\langle M\rangle), \qquad \theta_{k+1} := \mathsf{Pr}\big(\langle M\rangle, \langle\theta_k\rangle\big).$$

Canonical example. The first three terms are

$$\begin{aligned}
\theta_0 &= \mathsf{ATP}(\langle M\rangle),\\
\theta_1 &= \mathsf{Pr}\big(\langle M\rangle, \langle \mathsf{ATP}(\langle M\rangle)\rangle\big),\\
\theta_2 &= \mathsf{Pr}\Big(\langle M\rangle, \langle \mathsf{Pr}(\langle M\rangle, \langle \mathsf{ATP}(\langle M\rangle)\rangle)\rangle\Big).
\end{aligned}$$

In words: θ_0 says "I am an ATP," θ_1 says "I prove that I am an ATP," and θ_2 says "I prove that I prove that I am an ATP."

Proposition 15.12. *Every member of the reflection sequence of M is a self-descriptive formula in the sense of Definition 13.5.*

Proof. Induction on k. For $k = 0$ the formula is $\mathsf{ATP}(\langle M\rangle)$, whose intended subject is $\langle M\rangle$. Suppose θ_k is self-descriptive. Then θ_{k+1} is $\mathsf{Pr}(\langle M\rangle, \langle\theta_k\rangle)$, whose first argument is the tag $\langle M\rangle$ and whose intended subject is therefore again $\langle M\rangle$: it ascribes to $\langle M\rangle$ the property of proving a designated formula. □

Copyright © 2026 Brian Tenneson. All Rights Reserved.

Corollary 15.13. *If $M \vdash \theta_k$ then $\theta_k \in \mathrm{Self}(M)$.*

Proof. Immediate from Proposition 15.12 and Definition 13.9. □

Proposition 15.14. *The formulas $\theta_0, \theta_1, \theta_2, \ldots$ are pairwise distinct.*

Proof. We show by strong induction on $j + k$ that $\theta_j = \theta_k$ implies $j = k$.

If $j = k = 0$ there is nothing to prove. If exactly one of j, k is 0, say $j = 0 < k$, then θ_0 has principal predicate ATP while θ_k has principal predicate Pr, and these are distinct symbols of L; so $\theta_j \neq \theta_k$ and the implication holds vacuously.

If $j, k \geq 1$, write $j = j' + 1$ and $k = k' + 1$. From $\mathsf{Pr}(\langle M\rangle, \langle\theta_{j'}\rangle) = \mathsf{Pr}(\langle M\rangle, \langle\theta_{k'}\rangle)$ we get $\langle\theta_{j'}\rangle = \langle\theta_{k'}\rangle$, hence $\theta_{j'} = \theta_{k'}$ by injectivity of the naming operator (Definition 15.5). Since $j' + k' < j + k$, the inductive hypothesis gives $j' = k'$, hence $j = k$. □

15.5 Level n

Definition 15.15 (Level-n formal self-awareness)**.** Let $n \geq 1$. A tagged ATP M has *Level-n formal self-awareness* if $M \vdash \theta_{n-1}$, where $(\theta_k)_{k\geq 0}$ is the reflection sequence of M.

Remark 15.16. For $n = 1$ this is exactly Definition 15.3, since $\theta_0 = \mathsf{ATP}(\langle M\rangle)$. So Definition 15.15 extends the Version 12 scale rather than replacing it.

Definition 15.17 (Level set and level function)**.** The *level set* of a tagged ATP M is

$$L(M) := \{n \geq 1 : M \vdash \theta_{n-1}\},$$

and the *level* of M is

$$\mathrm{lev}(M) := \sup L(M) \in \{0, 1, 2, \ldots\} \cup \{\omega\},$$

with the convention $\sup \emptyset = 0$.

Proposition 15.18. $\mathrm{lev}(M) = 0$ *if and only if M has Level-0 formal self-awareness in the sense of Definition 15.1 restricted to reflection formulas; that is, if and only if M proves no member of its own reflection sequence.*

Proof. $\mathrm{lev}(M) = 0$ iff $L(M) = \emptyset$ iff there is no k with $M \vdash \theta_k$. □

15.6 Cumulativity Fails Without Hypotheses

One might expect the levels to be nested, so that Level-$(n+1)$ implies Level-n. They are not, and the failure is instructive.

Example 15.19 (A machine at Level 2 but not Level 1)**.** Let L contain the name $\langle M\rangle$, the unary predicate ATP, the binary predicate Pr, and names for formulas. Put

$$\theta_0 = \mathsf{ATP}(\langle M\rangle), \qquad \theta_1 = \mathsf{Pr}(\langle M\rangle, \langle\theta_0\rangle),$$

and let $Y = \{\theta_0, \theta_1\}$. Let the axiom base be $A = \{\theta_1\}$ and define $M = (C, H)$ over Y by $C(\Gamma, m) = \Gamma$ for all $\Gamma \subseteq Y$ and $m \in \mathbf{N}^+$, and $H(\Gamma, m) = 1$ iff $\Gamma \neq \emptyset$. The verification that M is a SIC is identical to Proposition 14.1.

On input A the machine halts with output $\{\theta_1\}$. Hence $M \vdash \theta_1$ but $M \nvdash \theta_0$. So M has Level 2 and not Level 1, and $L(M) = \{2\}$.

Copyright © 2026 Brian Tenneson. All Rights Reserved.